M. M. Institute of Computer Technology & Business Management (MCA), M.M. (Deemed To Be University), Mullana				Course Coverage Plan	
Session: 2025-26 (Even)			**Class/Semester: BCA – 6th Semester**		
Subject Code: BCA-608			**Subject Name: Data Visualization**		
Lecture: 3			**Tutorial: 0**		
Max. Marks Theory: 60			**Max. Marks Sessional: 40**	**Credit 3.0**	
Section	**Lecture No.**	**Date**	**Topic Discussed**	**Referred Book**	**Mode of Delivery (Regular/ICT)**
A	1	**06-01-26**	Introduction: Context of data visualization	**1**	Regular
	2	**07-01-26**	Definition, Methodology	**1,2**	Regular
	3	**09-01-26**	Visualization design objectives	**1,2**	Regular
	4	**13-01-26**	Key Factors – Purpose,	**1,2**	Regular
	5	**16-01-26**	Visualization function and tone, visualization design options	**1,2**	Regular
	6	**20-01-26**	Data representation, Data Presentation	**1,2**	ICT
	7	**21-01-26**	Seven stages of data visualization, widgets, data visualization tools	**1,2,3**	Regular
	8	**23-01-26**	Mapping-Time Series-Connections and Correlations	**1,2,3**	Regular
	9	**27-01-26**	Scatter plot Maps-Trees,	**1,2,3**	ICT
	10	**28-01-26**	Hierarchies, and Recursion –Networks and Graphs	**1,2,3**	Regular
B	1	**03-02-26**	Visualization techniques for time-series	**1,3**	ICT
	2	**04-02-26**	Mapping-Time series- Connections and correlations	**1,3**	Regular
	3	**06-02-26**	Indicator-Area Chart-Pivot table-Scatter charts,	**1,3**	Regular
	4	**06-02-26**	Scatter maps Trees & Graphs: Tree maps, Space filling	**1,3**	ICT
	5	**10-02-26**	Non-space filling methods-Hierarchies	**1,3**	Regular
	6	**10-02-26**	Recursion-Networks and Graphs	**1,2,3**	Regular
	7	**11-02-26**	Displaying Arbitrary Graphs-node	**1,2,3**	Regular
	8	**11-02-26**	Link graph-Matrix representation	**1,2,3**	Regular
	9	**13-02-26**	Graphs-Info graphics	**1,2,3**	Regular
	10	**13-02-26**	Revision	**1,2,3**	Regular
C	1	**17-02-26**	Time- Series data visualization	**1,2,3**	Regular
	2	**18-02-26**	Text data visualization	**1,2,3**	Regular
	3	**20-02-26**	Multivariate data visualization	**1,2,3**	Regular
	4	**24-02-26**	Case studies of visualization	**1,2,3**	Regular
	5	**25-02-26**	Processing streaming data for	**1,3**	Regular

			visualization		
	6	**27-02-26**	Presenting streaming data	**1,3**	Regular
	7	**03-03-26**	Streaming visualization techniques	**1,3**	Regular
	8	**06-03-26**	Streaming analysis	**1,3**	Regular
	9	**10-03-26**	Best practices of Data Streaming	**1,3**	Regular
	10	**11-03-26**	Revision	**1,2,3**	Regular
D	1	**13-03-26**	Security in Data Visualization	**3**	Regular
	2	**17-03-26**	Ports can visualize	**1,2**	Regular
	3	**18-03-26**	Vulnerability assessment and exploitation	**1,2**	Regular
	4	**20-03-26**	Firewall log visualization	**1,2**	Regular
	5	**25-03-26**	Intrusion detection log visualization	**1,2**	Regular
	6	**27-03-26**	Attacking and defending visualization Systems	**1,2**	Regular
	7	**01-04-26**	Creating secured visualization system	**1,2,3**	Regular
	8	**07-04-26**	Defending visualization Systems	**1,2,3**	Regular
	9	**08-04-26**	Security in Data Visualization	**1,2,3**	Regular
	10	**10-04-26**	Exploitation	**1,2,3**	Regular
	11	**15-04-26**	Revision	**1,2**	Regular
	12	**17-04-26**	**Test**	**1,3**	Regular

Text Books:

1. Tamara Munzer, Visualization Analysis and Design, CRC Press 2014.
2. Aragues, Anthony. Visualizing Streaming Data: Interactive Analysis Beyond Static Limits O'Reille Media, Inc., 2018
3. Christian Toninski, Heidrun Schumann, Interactive Visual Data Analysis, CRC press publication 2020.

Reference Books:

1. Robert Spence, Information Visualization: An Introduction, Third Edition, Pearson Education, 2014.
2. Joerg Osarek, Virtual Reality Analytics, Gordon's Arcade, 2016.
3. Matthew O. Ward, George Grinstein, Daniel Keim, Interactive Data Visualization: Foundation, Techniques and Applications", Second Edition, A. K. Peters/CRC Press, 2015.
4. Colin Ware, Information Visualization Perception for Design, 3rd ed., Margon Kaufmann Publishers, 2012.